$$\left.\begin{aligned} X_a &= \frac{1}{2}v_0 t + \frac{v_0}{2\omega_0}\sin\omega_0 t - \frac{l}{2} \\ X_b &= \frac{1}{2}v_0 t - \frac{v_0}{2\omega_0}\sin\omega_0 t + \frac{l}{2} \\ X_c &= \frac{1}{2}v_0 t + \frac{v_0}{2\omega_0}\sin\omega_0 t + \frac{l}{2} \\ X_d &= \frac{1}{2}v_0 t - \frac{v_0}{2\omega_0}\sin\omega_0 t + \frac{3l}{2} \end{aligned}\right\} \quad (7)$$

$$\left.\begin{aligned} v_a &= \frac{v_0}{2} + \frac{v_0}{2}\cos\omega_0 t \\ v_b &= \frac{v_0}{2} - \frac{v_0}{2}\cos\omega_0 t \\ v_c &= \frac{v_0}{2} + \frac{v_0}{2}\cos\omega_0 t \\ v_d &= \frac{v_0}{2} - \frac{v_0}{2}\cos\omega_0 t \end{aligned}\right\} \quad (8)$$

If b、c meet again (the second time), then

$$X_b = X_c$$

From (7),

$$\sin\omega_0 t = 0 \quad (9)$$

Least positive nonzero t:

$$\omega_0 t = \pi \quad (10)$$

b、c center-frame velocities at meeting, by (2), (9):

$$v_b'(\text{ft}) = \frac{v_0}{2}$$

$$v_c'(\text{at}) = -\frac{v_0}{2}$$

b、c meet: b right, c left; b、c collide again.

Just after collision they exchange velocities:

$$v_b'(\text{aft}) = -\frac{v_0}{2}$$

$$v_c'(\text{aft}) = \frac{v_0}{2}$$

Second b、c meeting: a、b、c、d shifts from equilibria follow from (10) and (1):

$$x_a' = 0$$

$$x_b' = 0$$

$$x_c' = 0$$

$$x_d' = 0$$

Thus both springs are natural and undeformed.

After second impact, table velocities of a、b、c、d: